

ESS and Generalized Leibniz Rule*

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July 1, 2025

Abstract

We introduce the extension spectral sequence, and explain the relationship between the differentials in ESS and extension problem. Using the extension spectral sequence and the theory of synthetic spectra, we may interpret the generalized Leibniz rule as a commutative square.

It is known that Adams SS is an effective tool for computing homotopy groups. An extension problem is to determine the group structure of a filtered group by knowing its associated graded groups. For instance, assume a filtered group $0 = F_2G \subset F_1G \subset F_0G = G$ with associated group equivalent to

$$\mathbb{Z}/2 \oplus \mathbb{Z}/2.$$

So, it is unclear that whether G is $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ or $\mathbb{Z}/4$. Unless, we know that $\exists x \in G$, such that $2x \neq 0$. In the setting of homotopy groups, this would lead to following question.

Question 0.1. Let X be a spectrum. Suppose $x \in \pi_*X$, is the image of x under the map $2 : X \rightarrow X$ non-zero?

Fact 0.2 (Adams). *For any spectrum $X \simeq X_p^\wedge$, the homotopy groups $\pi_*(X_p^\wedge)$ are filtered groups. The filtration is called Adams filtration, concretely, $\dots \subseteq F_1\pi_{t-s}X \subseteq F_0\pi_{t-s}X$. This filtration is functorial. The associated graded group is $\text{gr}_s F_\bullet \pi_{t-s}X = F_s\pi_{t-s}X / F_{s+1}\pi_{t-s}X \cong E_\infty^{s,t}(X)$. Here, the right hand side is the E_∞ -page of the Adams SS of X .*

Remark 0.3. Let $E = \mathbb{F}_p$, $\overline{E}$ be the fiber of the unit map $\mathbb{S} \rightarrow \mathbb{F}_p$. Then a canonical Adams tower of X is given by

$$\dots \rightarrow X_s \rightarrow X_{s-1} \rightarrow \dots \rightarrow X_1 \rightarrow X_0 = X.$$

Here, $X_s = \overline{E}^{\otimes s} \otimes X$. The Adams filtration is given by

$$\dots \rightarrow F_s\pi_*X \rightarrow F_{s-1}\pi_*X \rightarrow \dots \rightarrow F_1\pi_*X \rightarrow F_0\pi_*X.$$

Here, $F_s\pi_*(X) = \text{im}(\pi_*X_s)$.

Suppose we have a map of spectra $f : X \rightarrow Y$. Using the above fact, we can easily form the following filtered chain complex.

$$0 \longrightarrow F_\bullet \pi_{t-s}X \xrightarrow{F_\bullet \pi_{t-s}f} F_\bullet \pi_{t-s}Y \longrightarrow 0$$

The spectral sequence derived by the filtration is called f -extension spectral sequence (denoted f -ESS). The spectral sequence has the following signature:

$${}^f E_0^{s,t} \cong \text{gr } F_\bullet \pi_{t-s}X \oplus \text{gr } F_\bullet \pi_{t-s}Y \implies \ker(\pi_{t-s}f) \oplus \text{coker}(\pi_{t-s}f).$$

The differential has the form

$$d_n^f : {}^f E_n^{s,t} \rightarrow {}^f E_n^{s+n,t+n}.$$

*In this note, we refer to spectral sequence as "SS", Adams spectral sequence as "ASS", extension spectral sequence as "ESS".

Convention 0.4. Let ${}^f Z_n^{s,t} \subset E_\infty^{s,t}(X)$ be the subgroup consisting of elements for which the f -extension differentials $d_0^f, \dots, d_n^f$ vanish. Let ${}^f B_n^{s,t}(Y) \subset E_\infty^{s,t}(Y)$ be the subgroup generated by the sum of images of the f -extension differentials $d_0^f, \dots, d_n^f$. We define ${}^f B_{-1}^{s,t}(Y) = 0$ and ${}^f Z_{-1}^{s,t}(X) = E_\infty^{s,t}(X)$.

By definition we have

$${}^f E_n^{s,t} \cong {}^f Z_{n-1}^{s,t}(X) \oplus E_\infty^{s,t}(Y) / {}^f B_{n-1}^{s,t}(Y).$$

A nonzero d_n^f differential takes the form

$$d_n^f : {}^f Z_{n-1}^{s,t}(X) \rightarrow {}^f E_\infty^{s+n,t+n}(Y) / {}^f B_{n-1}^{s+n,t+n}(Y).$$

When we write

$$d_n^f(x) = y$$

for $x \in E_\infty^{s,t}(X)$ and $y \in E_\infty^{s+n,t+n}(Y)$, it is understood that x belongs to ${}^f Z_{n-1}^{s,t}(X)$, and

$$d_n^f(x) = y + {}^f B_{n-1}^{s+n,t+n}(Y).$$

If moreover, $y \notin {}^f B_{n-1}^{s+n,t+n}(Y)$, then this is an essential f -extension from x to y . Recall that a map f has Adams filtration $AF(f) \geq k$, if there exists $f_1, \dots, f_k$ each inducing a trivial map in $H\mathbb{F}_p$ -homology, such that f is homotopic to the composition $f_k \circ \dots \circ f_1$. The precise definition of Adams filtration is as follows.

Definition 0.5. A map $f : X \rightarrow Y$ has $AF(f) \geq k$, if there is a lift $f_k : X \rightarrow Y_k$ such that f is homotopic to the composition of f_k and the natural map $Y_k \rightarrow Y$.

Convention 0.6. For $x \in E_\infty^{s,t}(X)$, we denote $\{x\}$ to be the set of preimages of x under the map $F_s \pi_{t-s} X \rightarrow E_\infty^{s,t}(X)$. We use $[x]$ to refer to a specific element or a general element in $\{x\}$, with the choice understood from the context.

Definition 0.7. 1. For $x \in E_\infty^{s,t}(X)$, we say that an f -extension $d_n^f(x) = y$ has a crossing that hits Adams filtration p for some $p \leq s+n$, if there exists an element $x' \in E_\infty^{s+a,t+a}(X)$ with $a > 0$ and an f -extension $d_m^f(x') = y'$ for $0 \neq y' \in E_\infty^{s+a,t+a}(Y)$ such that

$$p = AF(y') = s + a + m \leq AF(y) = s + n.$$

2. We say that an f -extension $d_n^f(x) = y$ has no crossing if it has no crossing that hits the range $AF \geq AF(x) + 1$.

Proposition 0.8. An f -extension from x to y has no crossing in the range $AF \geq p$ iff for all $[x] \in \{x\}$ such that $AF(f(x)) \geq p$ we have $f([x]) \in \{y\}$. In particular, we have

1. An f -extension from x to y has no crossing iff for all $[x] \in \{x\}$ we have $f([x]) \in \{y\}$.
2. An f -extension $d_n^f(x) = 0$ has no crossing iff for all $[x] \in \{x\}$, we have

$$AF(f[x]) > AF(x) + n.$$

3. If $AF(f) = n$, then all d_n^f -differentials have no crossing in the f -ESS.

Proposition 0.9. Consider a homotopy commutative diagram of spectra

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ p \downarrow & & \downarrow q \\ Z & \xrightarrow{g} & W \end{array}$$

Suppose $m, n, l \geq 0$, $0 < k \leq m + l - n$,

$$\begin{aligned} x &\in E_\infty^{s,t}(X) & y &\in E_\infty^{s+n,t+n}(Y), \\ z &\in E_\infty^{s+m,t+m}(Z) & w &\in E_\infty^{s+m+l,t+m+l}(W), \end{aligned}$$

and

1. $d_n^f(x) = y$,
2. $d_m^p(x) = z$,
3. the differential in 1 or the differential in 2 has no crossing,
4. $d_l^g(z) = w$, and has no crossing that hits the range $AF \geq s + n + k$,
5. $d_{k-1}^q(y) = 0$, and has no crossing.

Then $d_{m+l-n}^q(y) = w$.

Proof. Condition 5 and Proposition 0.8 (2), we know that $AF(q[y]) \geq AF([y]) + k$. And $AF([y]) \geq s + n$. Therefore, $AF(q[y]) \geq s + n + k$. And by Condition 1, 2, 3, we can take an $[x]$, such that $f[x] \in \{y\}, p[x] \in \{z\}$. Therefore $AF(gp[x]) = AF(qf[x]) \geq s + n + k$. Then by Condition 4, we have for all $[z] \in \{z\}$ $g[z] \in \{w\}$. In particular, there exists $[x] \in \{x\}$ $p[x] \in \{z\}$, so we have $gp[x] \in \{w\}$. So, $qf[x] \in \{w\}$. Since $f[x] \in \{y\}$, this yields a q -extension from y to w . $\square$

Remark 0.10. The above Proposition has a synthetic version, i.e. all spectra appear in the diagram should be replaced by $\mathbb{F}_p$ -synthetic spectra. And the extension spectral sequence can be also generalized to the setting of synthetic spectra. Note that the differentials in the synthetic extension spectral sequence do not change the third degree (weight). Meaning that the synthetic f -ESS has the form:

$${}^f E_0^{s,t,w} \cong E_\infty^{s,t,w}(X) \oplus E_\infty^{s,t,w}(Y) \implies \ker(\pi_{t-s,w} f) \oplus \operatorname{coker}(\pi_{t-s,w} f),$$

$$d_n^f : {}^f E_n^{s,t,w} \rightarrow E_n^{s+n,t+n,w}.$$

Note that by deleting the condition 5, one gets a weaker version of the above proposition. However, Condition 4 should be replaced by the following condition.

- 4'. $d_l^g(z) = w$, and has no crossing.

In order to prove Proposition 0.8, we need following lemma.

Proposition 0.11. Consider $f : X \rightarrow Y$, $x \in E_\infty^{s,t}(X)$, $y \in E_\infty^{s+n,t+n}(Y)$ and $y' \in E^{s+m,t+m}(Y)$ for $m, n \geq 0$.

1. There is an f -extension from x to y , i.e. $d_n^f(x) = y$ in f -ESS iff there is a class $[x] \in \{x\}$ such that $f[x] \in \{y\}$.
2. Suppose we have an f -extension from x to both y and y' .

- (a) If $m = n$, then $y - y' \in {}^f B^{s+n,t+n}(Y)$ and there exists an element $x' \in E^{s+a,t+a}(X)$ for $0 < a \leq n$ such that

$$d_{n-a}^f(x') = y - y'.$$

- (b) If $m > n$, then the f -extension from x to y is inessential.

Proof. The equation $d_n^f(x) = y$ says

1. There is an element $x_{n-1} \in E_\infty^{s+n-1,t+n-1}(X)$, such that $f(x) = d_n^f(x) = f(x_{n-1}) \in E_\infty^{s+n,t+n}(Y) / {}^f B_{n-1}^{s+n,t+n}(Y)$.
2. So, we have $f(x_{n-1}) - y = f(x')$, with $x' \in E_\infty^{s+n,t+n}(X)$. Then we can choose $[x']$, such that $f([x] - [x']) \in f(y)$.

We thus have proved 1 in the Proposition, and the second point follows from 1. $\square$

Now, we present the proof of Proposition 0.8.

Proof of Proposition 0.8. We only give the Proof of the first statement.

($\implies$). Suppose for the converse holds. The extension has no crossing, and there is an $[x] \in \{x\}$, such that $f[x] \notin \{y\}$. Assume that $f[x] \in \{y'\}$. This equivalent to there is an f -extension from x to y' . Suppose that $y' \in E_\infty^{s+m,t+m}(Y)$. By Proposition 0.11, if $m = n$, then

$$d_n^f(x) = y, d_{n-a}^f(x') = y - y'$$

is a crossing in $AF \geq AF(x) + 1$. For $m > n$, we can find an f -extension from x' to y , where $AF(x') > AF(x)$. This yields a crossing in the range $AF \geq AF(x) + 1$. Hence, either case will arrive at a contradiction.

($\Leftarrow$). Suppose for all $[x]$, we have $f[x] \in \{y\}$, and there is a crossing. Meaning that there is an x' has (strict) higher filtration than x , and an f -extension from x' to y' . Here, y' has lower filtration than y . Then $[x] + [x'] \in \{x\}$, so $f([x] + [x']) \in \{y\}$. However, $f([x] + [x']) = f([x]) + f([x'])$. If $AF(y') < AF(y)$, then $f([x] + [x']) \in \{y'\}$. If $AF(y') = AF(y)$, then $f([x] + [x']) \in \{y + y'\}$. Either case would case a contradiction. $\square$

1 Adams spectral sequence

Recall an canonical Adams tower of X is given by $\{X_s\}_{s \geq 0} = \{\overline{E}^{\otimes s} \otimes X\}_{s \geq 0}$, the associated graded is $\{K_s\}_{s \geq 0} = \{\overline{E}^{\otimes s} \otimes X \otimes E\}_{s \geq 0}$. Take $E = \mathbb{F}_p$, then $(\mathbb{F}_p, \Gamma) = (\mathbb{F}_p, \mathcal{A}_p)$ is a Hopf algebroid (in fact, a Hopf algebra). By Kenneth property, we have the following

$$\pi_* K_s \cong H_*(\overline{\mathbb{F}}_p, \mathbb{F}_p)^{\otimes s} \otimes_{\mathbb{F}_p} H_*(X, \mathbb{F}_p) \cong \overline{\mathcal{A}}_p^{\otimes s} \otimes_{\mathbb{F}_p} H_*(X, \mathbb{F}_p).$$

This is the cobar complex $C_\Gamma^*(\mathbb{F}_p, H_*(X, \mathbb{F}_p))$. In fact we also have $\pi_* K_s \cong \text{Hom}_{\mathcal{A}_p^*}(H^*(X, \mathbb{F}_p), \mathbb{F}_p)$. Thus, the Adams E_2 -page is

$$\text{Ext}_{\mathcal{A}_p}^{s,t}(\mathbb{F}_p, H_*(X, \mathbb{F}_p)) \Longrightarrow \pi_{t-s}(X_p^\wedge).$$

Here, the Ext is the extension functor for $(\mathbb{F}_p, \mathcal{A}_p)$ -comodules. And we also have

$$\text{Ext}_{\mathcal{A}_p^*}^{s,t}(H^*(X, \mathbb{F}_p), \mathbb{F}_p) \Longrightarrow \pi_{t-s}(X_p^\wedge).$$

Theorem 1.1. *For $p = 2$*

1. $\text{Ext}^0 = \mathbb{Z}/2$ generated by $1 \in \text{Ext}^{0,0}$.
2. Ext^1 is spanned by $\{h_i : i \geq 0\}$ with $h_i \in \text{Ext}^{1,2^i}$ represented by $[\xi_i^{2^i}]$.
3. (Adams) Ext^2 is spanned by $\{h_i h_j : 0 \leq i \leq j, j \neq i + 1\}$.
4. (Wang) Ext^3 is spanned by $h_i h_j h_k$, subject to the relations

$$h_i h_j = h_j h_i, h_i h_{i+1} = 0, \quad h_i h_{i+2}^2 = 0, h_i^2 h_{i+2} = h_{i+1}^3,$$

along with elements

$$c_i = \langle h_{i+1}, h_i, h_{i+2}^2 \rangle \in \text{Ext}^{3,11 \cdot 2^i}.$$

Theorem 1.2 (Browder). *There exist framed manifolds in dimension n of Kervaire invariant one iff*

1. $n = 2^{j+1} - 2$, and
2. the element h_j^2 in the Adams spectral sequence survives to the E_∞ -page.

Remark 1.3. It is known that h_i^2 survives to E_∞ -page, for $1 \leq i \leq 5$, by the work of Browder, Mahowald–Tangora, Barratt–Jones–Mahowald. Using the method of equivariant homotopy theory, M.Hill–M.Hopkins–D.Ravenel showed that h_i^2 does not survive to E_∞ -page for $i \geq 7$.

Theorem 1.4 (Lin–Wang–Xu). *The element h_6^2 survives to the E_∞ -page.*

2 Synthetic spectra and ASS

Fact 2.1. *The synthetic Adams spectral sequence for νX*

$$\text{syn} E_2^{s,t,w}(\nu X) \Longrightarrow \pi_{t-s,w}(\nu X)$$

has an E_2 -page

$$\text{syn} E_2^{*,*,*}(\nu X) \cong E_2^{*,*}(X) \otimes \mathbb{F}_2[\lambda]$$

with differentials

$$d_r : \text{syn} E_r^{s,t,w} \rightarrow \text{syn} E_r^{s+r,t+r-1,w}.$$

Fact 2.2 (see the work of Burklund–Hahn–Senger). *The $(\mathbb{F}_2-)$ synthetic Adams spectral sequence for νX is isomorphic to the λ -Bockstein spectral sequence. More precisely, we have*

1.

$$E_2^{s,t}(X) \cong \pi_{t-s,t}(\nu X/\lambda).$$

2. *If we have a classical Adams spectral differential $d_r x = y$ for $x \in E_2^{s,t}(X) \cong \pi_{t-s,t}(\nu X/\lambda)$, then x admits a lift to $\pi_{t-s,t}(\nu X/\lambda^{r-1})$ whose image under the Bockstein*

$$\nu X/\lambda^{r-1} \rightarrow \Sigma^{1,-r+1}\nu X/\lambda$$

is equal to $d_r(x)$.

Fact 2.3. *The E_∞ -page of Adams spectral sequence for $\nu X/\lambda^r$ for $r \geq 2$ is given by*

$$E_\infty^{s,t,w}(\nu X/\lambda^r) \cong \begin{cases} Z_{r-t+w}^{s,t}(X)/B_{1+t-w}^{s,t}(X), & \text{if } 0 \leq t-w < r, \\ 0, & \text{otherwise.} \end{cases}$$

Fact 2.4. *Suppose in the ASS of X we have $d_r(x) = y$, where $x \in Z_{r-1}^{s,t}(X)$ and $y \in Z_\infty^{s+r,t+r-1}(X)$. Consider the map*

$$\delta : \nu X/\lambda^n \rightarrow \Sigma^{1,-n}\nu X/\lambda^{m-n}.$$

1. *If $r \geq n+1$, then we view x as an element of*

$$E_\infty^{s,t,t}(\nu X/\lambda^n) \cong Z_n^{s,t}(X),$$

and $\lambda^{r-n-1}y$ as an element of

$$E_\infty^{s+r,t+r-1,t+n}(\nu X/\lambda^{m-n}) \cong Z_{m-r+1}^{s+r,t+r-1}(X)/B_{r-n}^{s+r,t+r-1}(X).$$

We then have

$$d_r^\delta(x) = \lambda^{r-n-1}y,$$

which is trivial if $r > m$.

2. *If $r < n+1$, then we view $\lambda^{n+1-r}x$ as an element of*

$$E_\infty^{s,t,t-n+1+r}(\nu X/\lambda^n) \cong Z_r^{s,t}(X)/B_{n+2-r}^{s,t}(X)$$

and y as an element of

$$E_\infty^{s+r,t+r-1,t+r-1}(\nu X) \cong Z_\infty^{s+r,t+r-1}(X).$$

In this case, we have

$$d_r^\delta(\lambda^{n+1-r}x) = y.$$

3 Generalized Leibniz rule

Convention 3.1. We refer to Fact 2.4(= Proposition 4.6 in LWX24) and BHS23 A.1 as rigidity (theorem). We call Proposition 0.9 square theorem.

- For a map $f : X \rightarrow Y$, we set $\hat{f}$ satisfying $\nu f = \lambda^{e(f)}\hat{f}$.
- We denote the mod λ^{r-1} reduction map by $\hat{f}_{r-1}$:

$$\hat{f}_{r-1} : \Sigma^{0,e(f)}\nu X/\lambda^{r-1} \rightarrow \nu Y/\lambda^{r-1}.$$

Theorem 3.2. *Let $f : X \rightarrow Y$ be a map between two classical spectra. Suppose that $2 \leq n \leq r, e(f) \leq m \leq n-2+e(f), l \geq e(f)$, and we have*

$$\begin{aligned} x &\in Z_{r-1}^{s,t}(X), & y &\in Z_{r-1-m+e(f)}^{s+m,t+m}(Y) \\ x_\infty &\in Z_\infty^{s+r,t+r-1}(X), & y_\infty &\in Z_\infty^{s+r+l,t+r+l-1}(Y) \end{aligned}$$

and the following conditions hold:

1. $d_r(x) = x_\infty$,
2. $d_m^{f,E_n}(x) = y$,
3. $d_l^{f,E_\infty}(x_\infty) = y_\infty$,
4. the differential in (1) has no crossing on the E_n -page or (2) has no crossing.
5. the differential in (3) has no crossing.

Then we have an Adams differential

$$d_{r+l-m}(y) = y_\infty.$$

Proof. By Fact 2.4, condition 1, we can view $x \in E_\infty^{s,t,t}(\nu X/\lambda^{n-1}) \cong Z_{n-1}^{s,t}(X)$, and by rigidity we have $d_r^\delta(x) = \lambda^{r-(n-1)-1}x_\infty$. And by condition 2, 3, and Definition 5.4 in LWX24, we get following two equations.

$$\begin{aligned} d_m^{\hat{f}^{n-1}}(x) &= \lambda^{m-e(f)}y, \\ d_l^{\hat{f}}(x_\infty) &= \lambda^{l-e(f)}y_\infty. \end{aligned}$$

Since the differentials are λ -linear, we then have

$$d_l^{\hat{f}}(\lambda^{r-n}x_\infty) = \lambda^{r-n+l-e(f)}y_\infty.$$

Now, consider the following commutative diagram of synthetic spectra

$$\begin{array}{ccc} \Sigma^{0,e(f)}\nu X/\lambda^{n-1} & \xrightarrow{f} & \nu Y/\lambda^{n-1} \\ \downarrow \delta_X & & \downarrow \delta_Y \\ \Sigma^{1,-n+1+e(f)}\nu X & \xrightarrow{\hat{f}} & \Sigma^{1,-n+1}\nu Y \end{array}$$

Condition 4,5 ensure that no crossing conditions in Proposition 0.9 (square theorem) hold, thus we conclude that

$$d_{r+l-m}^{\delta_Y}(\lambda^{m-e(f)}y) = \lambda^{r+l-n-e(f)}y_\infty.$$

This is equivalent to the following equation by rigidity

$$d_{r+l-m}(y) = y_\infty.$$

□

Remark 3.3. In Chua's work, the proof overlooked the existence of lift of x satisfying both condition 1 and 2. This would require certain no crossing conditions, which appeared in Proposition 0.9.

Example 3.4. Now, set

$$n = 2, r = 2, s = 1, t = 35, m = 1, l = 2, e(f) = 1.$$

Consider the map $\nu : S^3 \rightarrow S^0$, and following elements.

$$\begin{aligned} x &= h_5 \in \text{Ext}_{\mathcal{A}}^{1,32} \cong Z_1^{1,35}(S^3), \\ y &= h_2 h_5 \in \text{Ext}_{\mathcal{A}}^{2,36} \cong Z^{2,36}(S^0), \\ x_\infty &= h_0 h_4^2 \in \text{Ext}_{\mathcal{A}}^{3,33} \cong Z_\infty^{3,36}(S^3), \\ y_\infty &= h_0 p \in \text{Ext}_{\mathcal{A}}^{5,38} \cong Z^{5,38}(S^0). \end{aligned}$$

- By the result of Adams, we know that $d_2(x) = x_\infty$.
- Because ν is detected by h_2 in the Adams E_2 -page, thus we have $d_1^{f,E_2}(x) = y \iff d_1^{\hat{f}^1}(x) = \lambda^{1-e(f)}y = y$. The last statement is equivalent to $[h_2] \circ [h_5] = [h_2 h_5]$ by Fact 2.2. This differential has no crossing by definition and degree reason.

- First, we have $d_2^{f,E_3}(x_\infty) = y_\infty$. By definition, the differential $d_2^{f,E_3}(h_0h_4^2) = h_0p$ is equivalent to the existence of an $[h_2]/\lambda^2$ -extension from $[h_0h_4^2]$ to $\lambda^{2-1}[h_0p]$. So it is equivalent to the existence of synthetic homotopy class $[h_0h_4] \in \pi_{30+3,33+4}(S^{3,4}/\lambda^2)$ such that

$$[h_0h_4^2] \cdot [h_0] = [\lambda h_0p] \in \pi_{33,37}S^{0,0}/\lambda^2.$$

This will be proved using generalized Mahowald trick and computer program. By checking certain no crossing condition (by generalized Mahowald trick), the differential $d_2^{f,E_3}(x_\infty) = y_\infty$ can be promoted to the differential $d_2^{f,E_\infty}(x_\infty) = y_\infty$.

- By definition (of a crossing differential), the differential $d_2(x) = x_\infty$ has no crossing. By degree reason, the only possible crossing for $d_2^{f,E_3}(x)$ is an (f, E_2) -extension from $h_0^2h_4^2$ to h_0p . This would implies that $h_0^2h_4^2 \cdot h_2 = 0 = h_0p$, a contradiction. So differential (3) has no crossing.
- We may thus conclude that $d_3(y) = y_\infty$, i.e. $d_3(h_2h_5) = h_0p$.

Example 3.5. Consider the map $f = 2 : S^0 \rightarrow S^0$, and the classical Hopf invariant one differential

$$d_2(h_4) = h_0h_3^2.$$

Set

$$n = 2, r = 2, s = 2, t = 16, m = 1, l = 2, e(f) = 1,$$

and

$$\begin{aligned} x &= h_4 \in \text{Ext}^{1,16}_1(S^0), y = h_0h_4 \in \text{Ext}^{2,17}_1(S^0). \\ x_\infty &= h_0h_3^2 \in \text{Ext}^{3,17}_\infty(S^0), y_\infty = 0 \in \text{Ext}^{5,19}_\infty(S^0). \end{aligned}$$

We have:

1. $d_2(x) = x_\infty$. (Adams' work).
2. $d_1^{f,E_2}(x) = y$, since the product of h_4 by h_0 is h_0h_4 in Ext .
3. $d_2^{f,E_\infty}(x_\infty) = 0$.
4. (a) by definition the differential (1) has no crossing.
(b) by definition (or degree reason) (2) has no crossing.
5. the differential in (3) has a crossing:

$$d_1^{f,E_\infty}(d_0) = h_0d_0.$$

By definition and rigidity, the differential (3) is equivalent to the existence of a synthetic homotopy class $[h_0h_3^2] \in \pi_{14,17}(S^{0,0})$, such that

$$[h_0h_3^2] \cdot [h_0] = 0 \in \pi_{14,18}(S^{0,0}).$$

We learn that $\pi_{14,18}(S^{0,0}) \cong \mathbb{Z}/2\{\lambda h_0d_0\}$, the generator $[\lambda h_0d_0]$ is an $[h_0]$ -multiple of a class $[\lambda d_0]$ has higher Adams filtration than $h_0h_3^2$.

1. If there is a class satisfying $[h_0h_3^2][h_0] = 0$, then we have a differential $d_2^{f,E_\infty}(x_\infty) = 0$.
2. If for any class $[h_0h_3^2]$ such that $[h_0h_3^2][h_0] = \lambda h_0d_0$, i.e. $d_2^{f,E_\infty}(x) = h_0d_0$. Then this is inessential. So, we can conclude there is a differential $d_2^{f,E_\infty}(x_\infty) = 0$.

If the no-crossing condition were removed, then we would arrive at a contradiction.

$$d_3(h_0h_4) = 0, d_3(h_0h_4) = h_0d_0.$$

Remark 3.6. Consider the map $\alpha = 2 : S^{0,0} \rightarrow S^{0,0}$, and elements

$$h_4 \in \pi_{*,*}(S^{0,0}/\lambda), d_2(h_4) = h_0h_3^2.$$

By the definition of Chua's work, and the discussion before, we know that

$$[h_0h_3^2][h_0] = 0$$

is a maximal α -extension. We thus conclude

$$d_3(h_3h_4) = 0,$$

which is incorrect.